HAT-P-36b

R-0728

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_260311` · created 2026-07-28T00:37:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461110.8411 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.011
Transit depth (Rp/Rs)^2	1.91 +/- 0.29 [%]
Orbital Inclination (inc)	86.72 +/- 2.14
Airmass coefficient 1 (a1)	1.0045 +/- 0.0099
Airmass coefficient 2 (a2)	-0.003 +/- 0.042
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#2 - [296, 98]
Optimal Aperture	3.43
Optimal Annulus	8.51
Transit Duration (day)	0.0954 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.011 vs 0.126 (deviation 1.09σ)
Duration fit vs published	0.0954 d vs 0.0925 d (deviation 0.72σ)
Mid-time O–C	5.9 min
Depth SNR	12.5
Residual scatter	0.99 %

Light curve

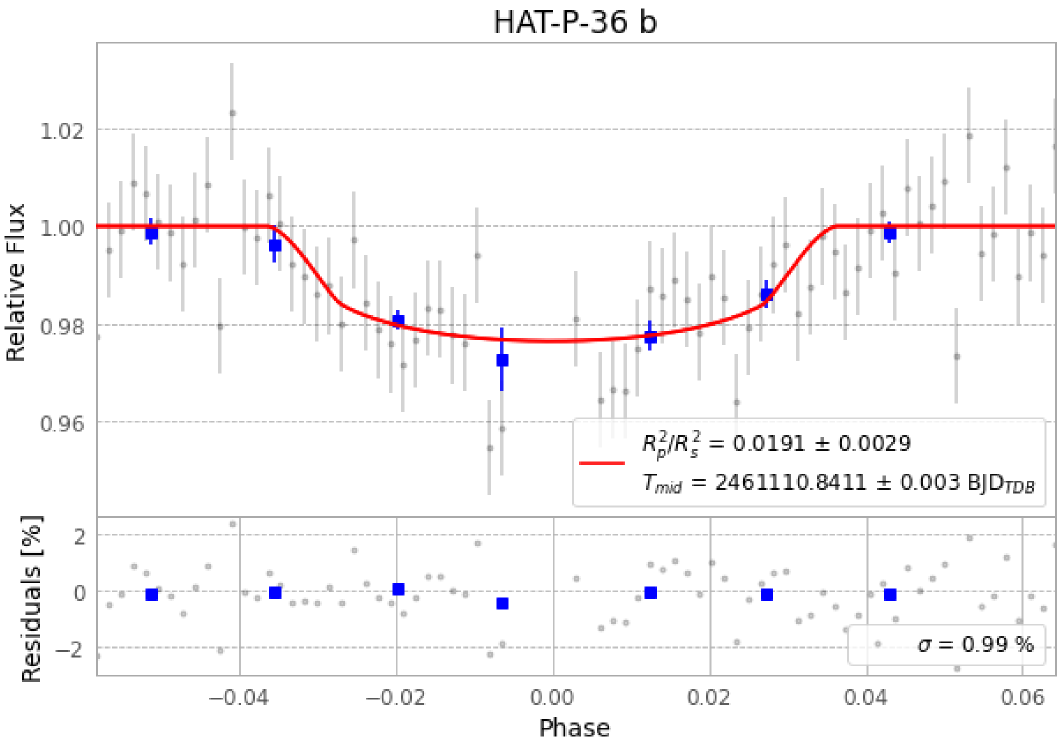

Figure 1 — FinalLightCurve_HAT-P-36 b_2026-03-10.png (SHA-256 43386687109b555e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 113], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7f574da369c82569…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

79 FITS frames (52 MB), `HATP-36260311061215.FITS` → `HATP-36260311100616.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260311.log (57fe19a62fb0…), FinalLightCurve_HAT-P-36 b_2026-03-10.png (43386687109b…), FinalLightCurve_HAT-P-36 b_2026-03-10.csv (16efdbd71fe1…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)